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Unit 2 - Understanding Sensors and Signals

PDF Documents

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This PDF document is included in the unit materials.

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UNIT 2 UNDERSTANDING SENSORS AND SIGNALS

In this unit, you will learn about sensors and why they are so important for keeping us safe. Sensors are like the playground's "eyes and ears." They can detect what is happening around them and help make smart decisions.

WHAT IS A SENSOR?

A sensor is a small device that can "sense" things like distance, light, or motion. It works by measuring something in the environment and sending information to another device, like a light or a speaker, to make something happen.

For example:

Motion sensors can tell when someone is moving. These are used in automatic doors at the grocery store, so the door opens when you walk up to it.

Light sensors can tell how bright or dark it is. Your phone might use one to adjust the screen brightness automatically.
Distance sensors, like the one we will use for the Smart Slide, measure how close or far away something is.

HOW DO DISTANCE SENSORS WORK?

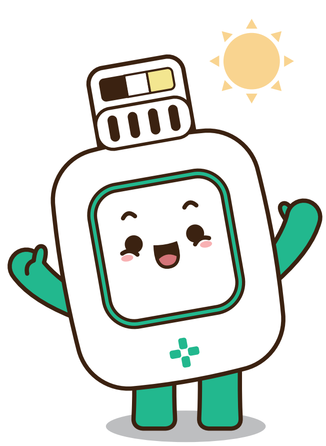
The distance sensor is a special type of sensor that measures how far away something is. It can detect when a person or object is close to the slide. This sensor works by sending out light or sound waves, which bounce back when they hit something. The sensor then measures how long it takes for the waves to return, which tells it how far away the object is.

SENSORS SEND SIGNALS

Once a sensor detects something, it sends a signal to another device, like a light or alarm, to make something happen. This is how our Smart Slide will work:

- If the sensor detects someone close to the bottom of the slide, it will send a signal to turn the LED light red, meaning "stop and wait."
- If no one is close, the sensor will send a signal to turn the LED light green, meaning "go ahead, it's safe to slide!"

This process helps the playground equipment "think" and keep everyone safe.

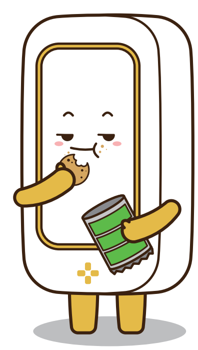


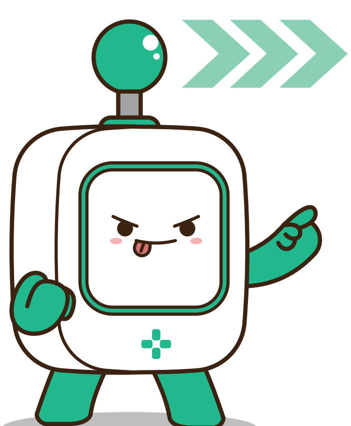
HOW DO DISTANCE SENSORS WORK?

Sensors are very important because they help us stay safe in many places. You may not realize it, but sensors are used all around you:

- In cars, distance sensors help drivers park safely by beeping when they get too close to another car.
- In homes, smoke detectors have sensors that detect smoke and set off an alarm if there is a fire.

Sensors are smart and can make decisions quickly, keeping us safe without us even noticing!





Now that you understand how sensors and signals work, you are ready for the next step—building your very own Smart Slide! In the next unit, you will learn how to set up a distance sensor and program it to control the LED light, making sure your slide is safe for everyone to use. Are you ready to get started?

Activity:

Imagine you are using a distance sensor to create your own safety system. What would it be? Draw your idea below and label the parts of your system. For example, it could be a Smart Door that only opens when no one is blocking the way.